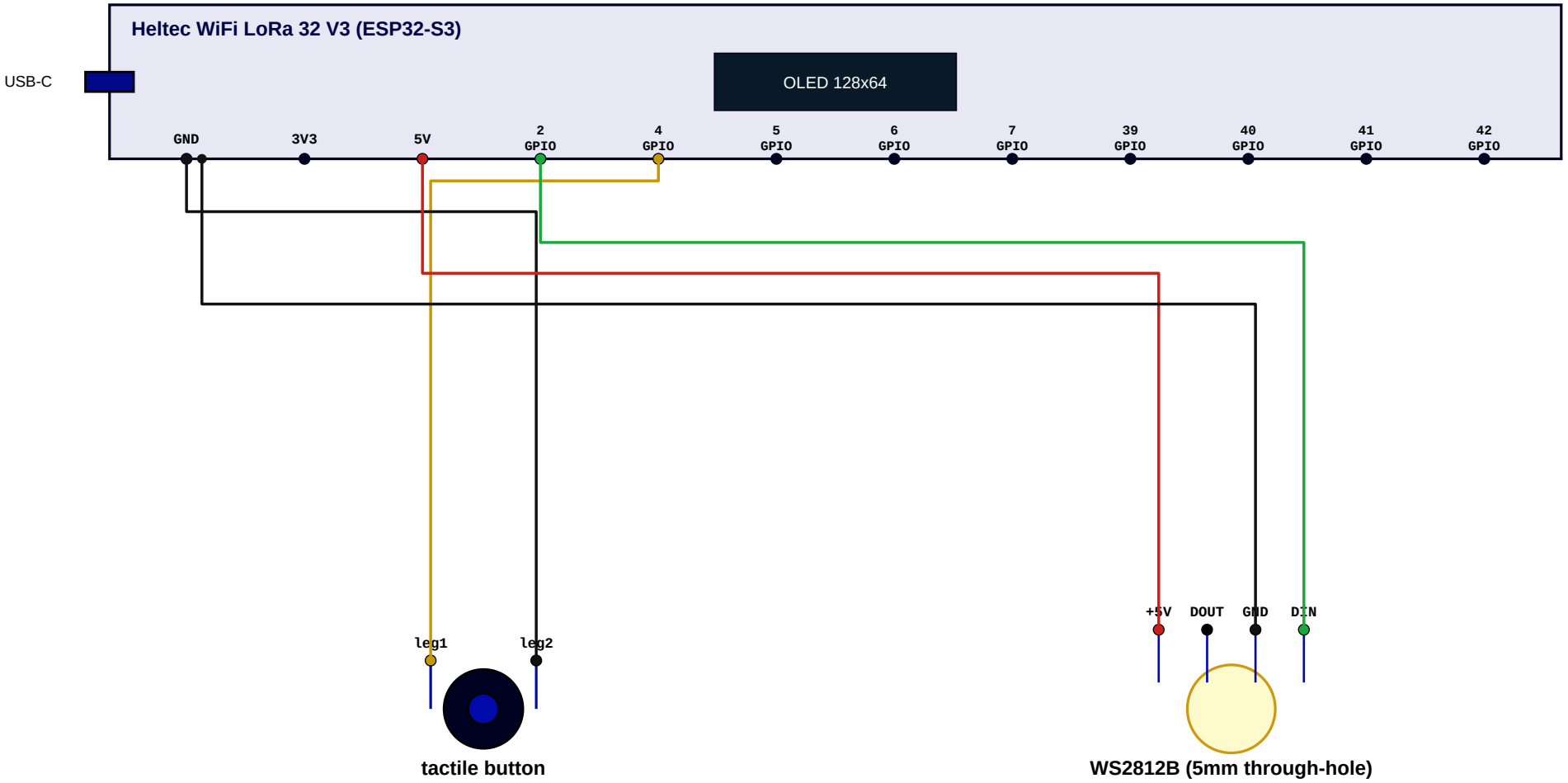
SO2R Box — POC Wiring Diagrams

One landscape page per milestone. Heltec pin labels along the bottom edge of the module match the pin map in `so2r_poc.ino`. Wires are colour-coded and use dedicated horizontal lanes (one lane per wire) so no two routes share a coordinate.

Milestone 1 - OLED + 1 button + 1 WS2812B

Toolchain sanity check. No external ICs. The WS2812B is driven direct from 3.3V GPIO; expect glitches (REVIEW B6 in person).



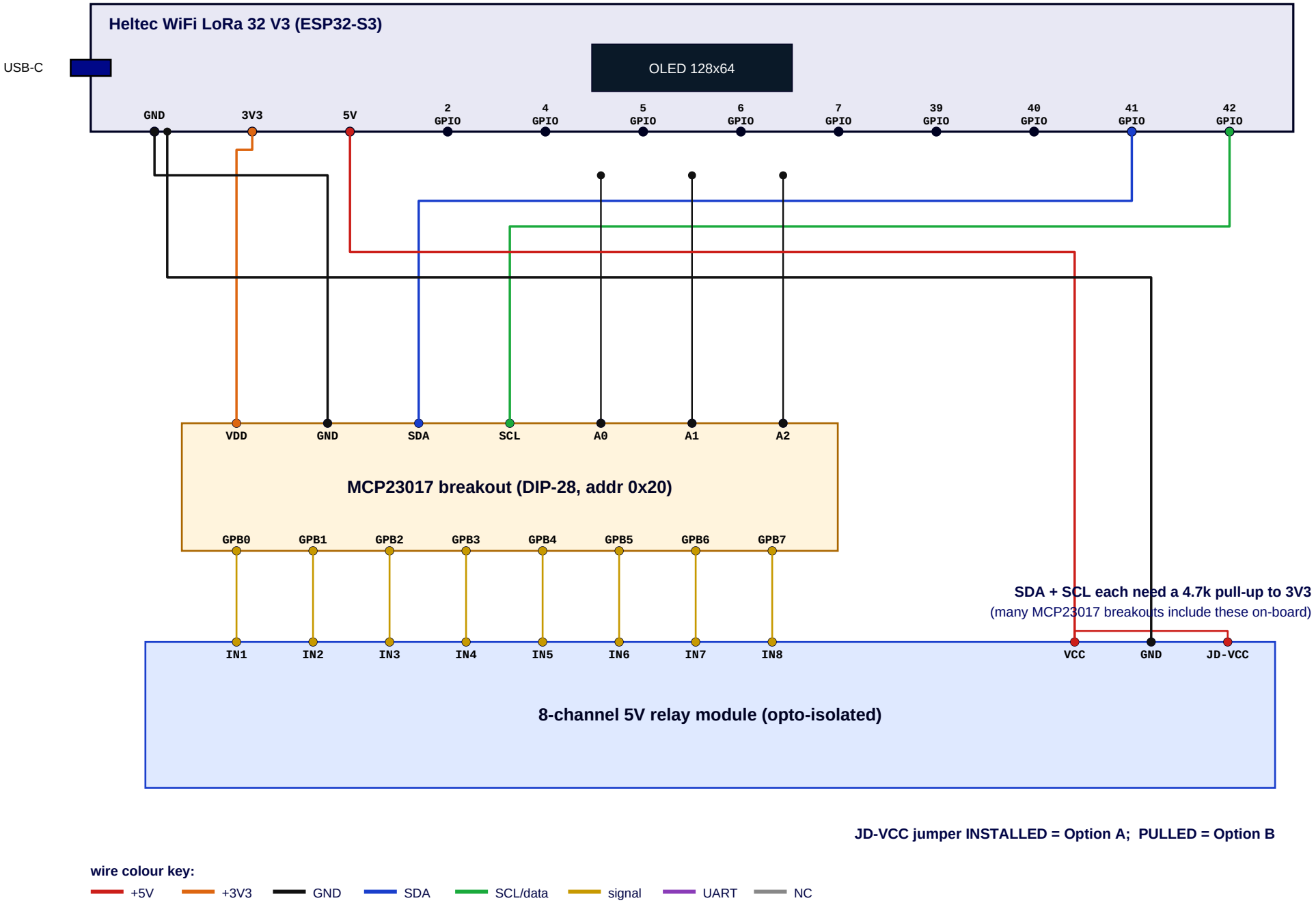
wire colour key:

+5V · +3V3 · GND · SDA · SCL/data · signal · UART · NC

From	To	Wire colour	Notes
`Heltec GPIO 4`	`Button leg 1`	yellow	INPUT_PULLUP; pressed = LOW.
`Button leg 2`	`Heltec GND`	black	Common ground.
`Heltec GPIO 2`	`WS2812B DIN`	green	Direct 3.3V drive - documented to glitch (REVIEW B6).
`Heltec 5V`	`WS2812B +5V`	red	Pin VDD on the LED.
`Heltec GND`	`WS2812B GND`	black	Pin GND on the LED.
`WS2812B DOUT`	`(no connection)`	grey	Chain-out; unused for one LED.

Milestone 2 - MCP23017 + 8-channel relay module

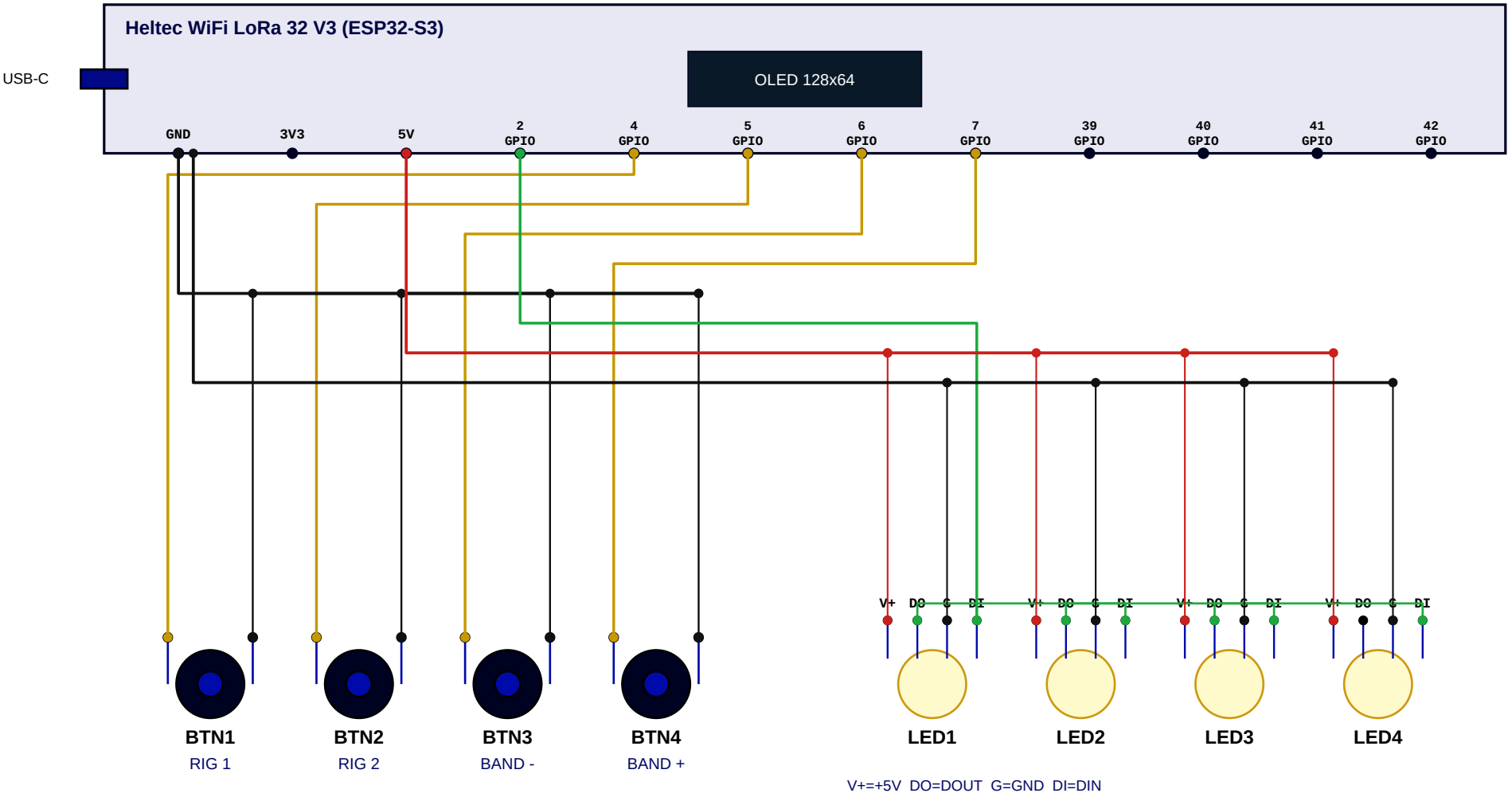
The architectural linchpin. MCP23017 sits on the SECONDARY I2C bus (GPIO 41 / 42) so it doesn't share with the OLED. All eight relay drives are straight vertical wires between MCP23017 port B and the relay module's IN1-IN8 header. Coil power = USB 5V (Option A); for the 60s stress test pull the JD-VCC jumper and feed it from a 5V USB charger (Option B).


Heltec WiFi LoRa 32 V3 (ESP32-S3)
USB-C
OLED 128x64
GND
3V3
5V
2 GPIO
4 GPIO
5 GPIO
6 GPIO
7 GPIO
39 GPIO
40 GPIO
41 GPIO
42 GPIO
VDD
GND
SDA
SCL
A0
A1
A2
MCP23017 breakout (DIP-28, addr 0x20)
GPB0
GPB1
GPB2
GPB3
GPB4
GPB5
GPB6
GPB7
IN1
IN2
IN3
IN4
IN5
IN6
IN7
IN8
VCC
GND
JD-VCC
8-channel 5V relay module (opto-isolated)
SDA + SCL each need a 4.7k pull-up to 3V3
(many MCP23017 breakouts include these on-board)
JD-VCC jumper INSTALLED = Option A; PULLED = Option B
wire colour key:
+5V
+3V3
GND
SDA
SCL/data
signal
UART
NC

From	To	Wire colour	Notes
`Heltec 3V3`	`MCP23017 VDD`	orange	Logic supply.
`Heltec GND`	`MCP23017 GND + Relay GND`	black	Single shared ground.
`Heltec GPIO 41`	`MCP23017 SDA`	blue	+ 4.7k pull-up to 3V3.
`Heltec GPIO 42`	`MCP23017 SCL`	green	+ 4.7k pull-up to 3V3.
`MCP23017 A0/A1/A2`	`MCP23017 GND (short stubs)`	black	Sets I2C address to 0x20.
`MCP23017 GPB0..GPB7`	`Relay IN1..IN8`	yellow	Eight straight vertical drops; firmware polarity-aware.
`Heltec 5V`	`Relay VCC + JD-VCC`	red	Option A: JD-VCC jumper installed.
`5V USB charger`	`Relay JD-VCC (jumper pulled)`	red	Option B for the stress test.

Milestone 3 - Panel UI (4 buttons + 4 WS2812B chain)

Adds the operator panel on top of M2's MCP23017 / relay wiring (which is unchanged). Four buttons feed GPIO 4-7 with INPUT_PULLUP and share a single GND bus. Four WS2812B LEDs daisy-chain DOUT-to-DIN with shared +5V and GND rails.



Milestone 2 wiring (MCP23017 + 8-channel relay module) stays in place. This page shows ONLY the panel additions.

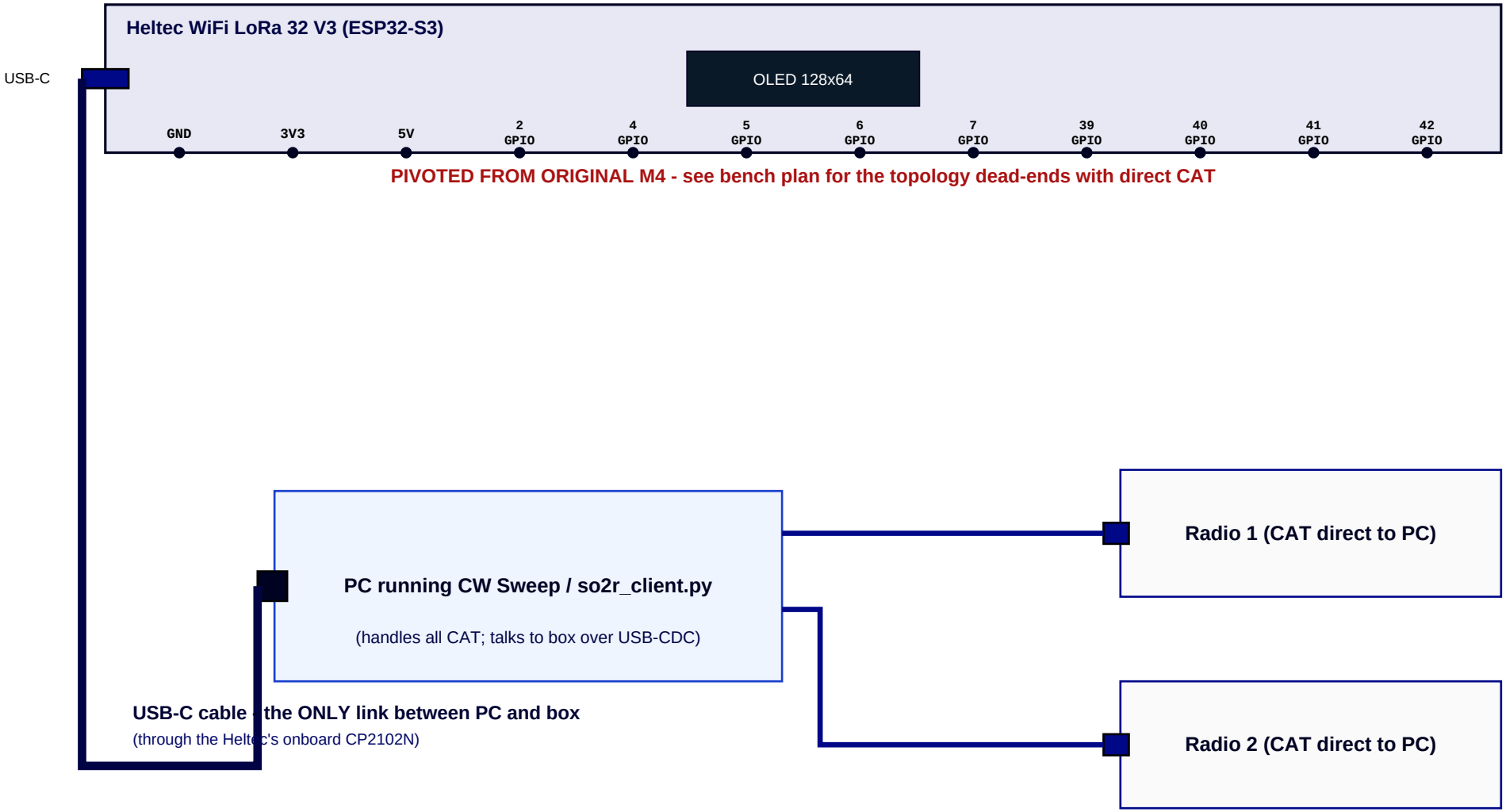
wire colour key:

+5V · +3V3 · GND · SDA · SCL/data · signal · UART · NC

From	To	Wire colour	Notes
`Heltec GPIO 4`	`BTN1 leg 1`	yellow	RIG 1 select.
`Heltec GPIO 5`	`BTN2 leg 1`	yellow	RIG 2 select.
`Heltec GPIO 6`	`BTN3 leg 1`	yellow	Band down.
`Heltec GPIO 7`	`BTN4 leg 1`	yellow	Band up.
`All button leg 2`	`Shared GND bus -> Heltec GND`	black	One bus, one wire to Heltec.
`Heltec GPIO 2`	`LED1 DIN`	green	Chain start.
`LEDn DOUT`	`LED(n+1) DIN`	green	Daisy-chain 3 hops.
`Heltec 5V`	`Shared LED +5V rail`	red	All four LEDs in parallel.
`Heltec GND`	`Shared LED GND rail`	black	All four LEDs in parallel.

Milestone 4 - PC <-> box USB-CDC protocol (pivoted)

Original M4 (ESP32 talks CAT directly to the radio via an external CP2102N) was abandoned on the bench. The radio's USB-CAT port is a USB device, not a host - two USB devices can't talk without an intermediate host. The new M4 puts all CAT on the PC and treats the box as a USB-CDC peripheral. NO external UART hardware on the box.



ARCHITECTURE: the box is a smart USB-CDC peripheral. The PC owns all CAT and tells the box which antenna / PTT / audio routing to apply. Panel button presses on the box are reported back to the PC as asynchronous events. No external UART hardware on the box; the Heltec's onboard CP2102N is the link.

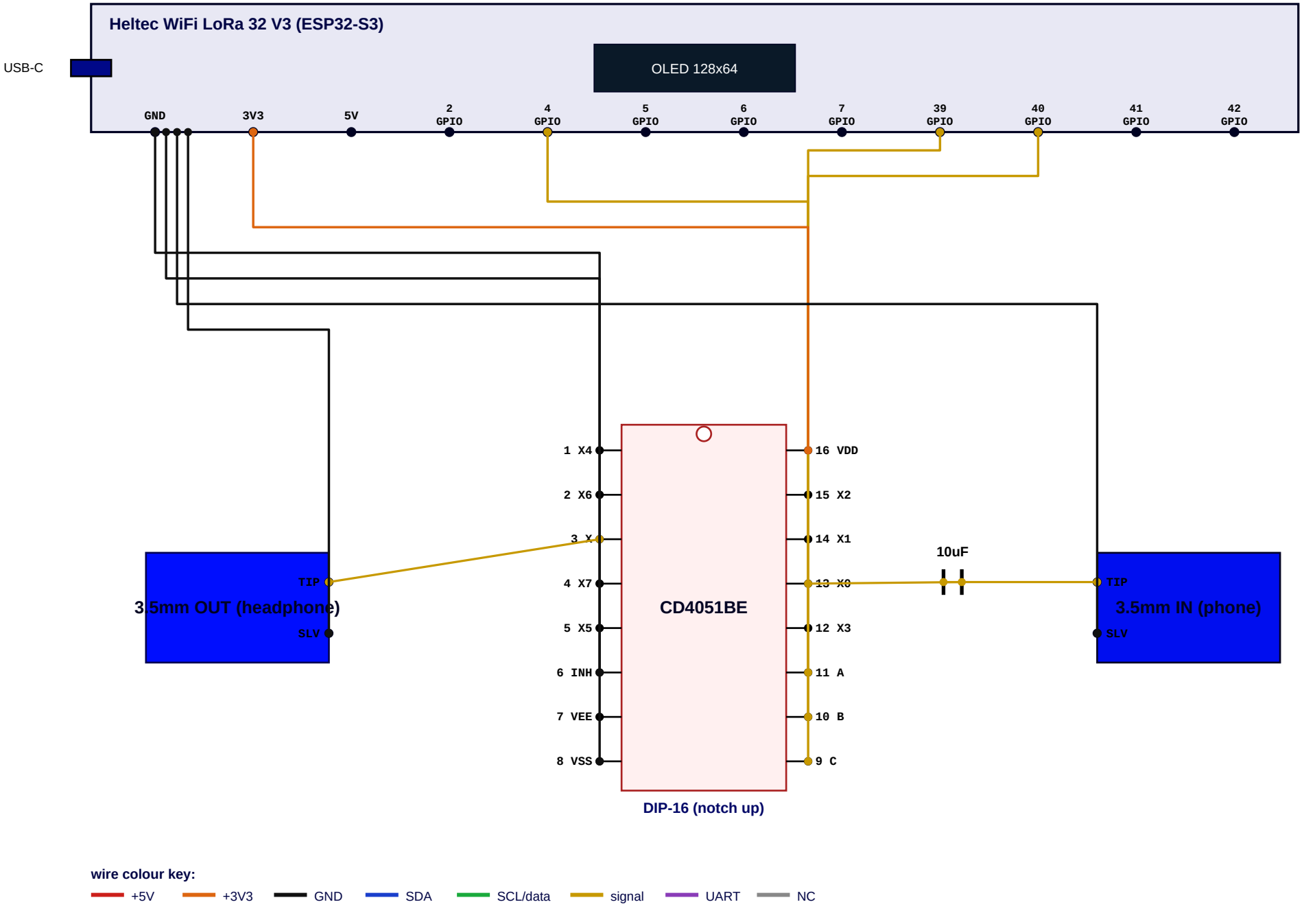
wire colour key:

+5V +3V3 GND SDA SCL/data signal UART NC

From	To	Wire colour	Notes
Heltec USB-C	Mac USB port	(USB cable)	Onboard CP2102N is the only PC link. Heltec enumerates as /dev/cu.usbserial-* on macOS.
Radio 1 CAT	Mac (separately)	(USB or serial)	Each radio's CAT cable goes to the PC, not the box. PC handles all per-vendor CAT (Hamlib / Sweep).
Radio 2 CAT	Mac (separately)	(USB or serial)	Same arrangement for the second radio.
(MCP23017 + relay module)	(unchanged from M2)	(prior wiring)	Antenna relay matrix wiring is identical to milestone 2.
External CP2102N module	(removed)	(none)	No longer in the runtime path. Keep it for future bench tools.

Milestone 5 - CD4051 audio mux (stretch)

One CD4051BE 8:1 analog mux. Phone audio enters via a 10 uF coupling cap on X0 (pin 13). Headphone hangs off X (pin 3). Three address GPIOs choose the channel. INH tied LOW = mux enabled. Single-supply operation - negative audio peaks will clip; fine for POC.


Heltec WiFi LoRa 32 V3 (ESP32-S3)
USB-C
OLED 128x64
GND
3V3
5V
2 GPIO
4 GPIO
5 GPIO
6 GPIO
7 GPIO
39 GPIO
40 GPIO
41 GPIO
42 GPIO
1 X4
2 X6
3 X
4 X7
5 X5
6 INH
7 VEE
8 VSS
16 VDD
15 X2
14 X1
13 X0
12 X3
11 A
10 B
9 C
CD4051BE
DIP-16 (notch up)
10uF
TIP
SLV
3.5mm OUT (headphone)
TIP
SLV
3.5mm IN (phone)
wire colour key:
+5V
+3V3
GND
SDA
SCL/data
signal
UART
NC

From	To	Wire colour	Notes
Phone audio TIP	10 uF cap -> CD4051 pin 13 (X0)	yellow	Single channel; either L or R.
Phone audio SLV	Heltec GND	black	Mandatory shared ground.
Heltec GPIO 39	CD4051 pin 11 (A)	yellow	Address LSB.
Heltec GPIO 40	CD4051 pin 10 (B)	yellow	Address mid.
Heltec GPIO 4	CD4051 pin 9 (C)	yellow	Address MSB.
Heltec 3V3	CD4051 pin 16 (VDD)	orange	Single-supply.
Heltec GND	CD4051 pin 8 (VSS)	black	Ground.
Heltec GND	CD4051 pin 6 (INH)	black	INH LOW = mux enabled.
CD4051 pin 3 (X)	Headphone TIP	yellow	Common output.
Headphone SLV	Heltec GND	black	Closes return path.